

# ETP story formalization · majority voting, frontier closed

## models, deepest-nesting pairs (bin 10), t0.7 vs t1.0

Stories generated from Equational Theories Project implications (storyform.py), formalized back by each model over OpenRouter, and graded syntactically (checkform.py). Correct = exact, or an accepted symmetry (sides swapped, or both equations uniformly dualized).

experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t0.7-t0-baseline · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t0.7-vote1 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t0.7-vote3 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t0.7-vote5 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t0.7-vote10 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t1-vote1 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t1-vote3 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t1-vote5 · experiments/12-majority-voting-frontier/runs/run-eqstrat20-bin10-s0-story-t1-vote10

|      |        |         |              |                 |
|------|--------|---------|--------------|-----------------|
| Runs | Models | Stories | Graded calls | Overall correct |
| 9    | 3      | 20      | 540          | 40%             |

20 pairs, 20 per per-equation operation bin 10–10, both laws from the same bin · seed 0 · temp0 single-pass vs t0.7 vote@1 vs t0.7 vote@3 vs t0.7 vote@5 vs t0.7 vote@10 vs t1.0 vote@1 vs t1.0 vote@3 vs t1.0 vote@5 vs t1.0 vote@10

FIGURE 1 Correct rate by model and condition — balanced 20

The same 20 stories under each condition; each row compares one model across all conditions. Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

- temp0 single-pass

t0.7 vote@1

t0.7 vote@3

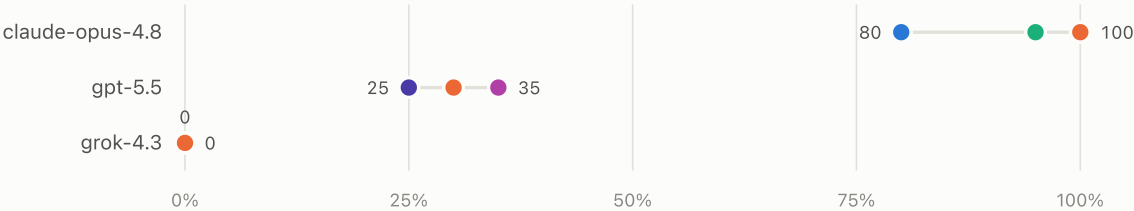
t0.7 vote@5

t0.7 vote@10

t1.0 vote@1
- t1.0 vote@3

t1.0 vote@5

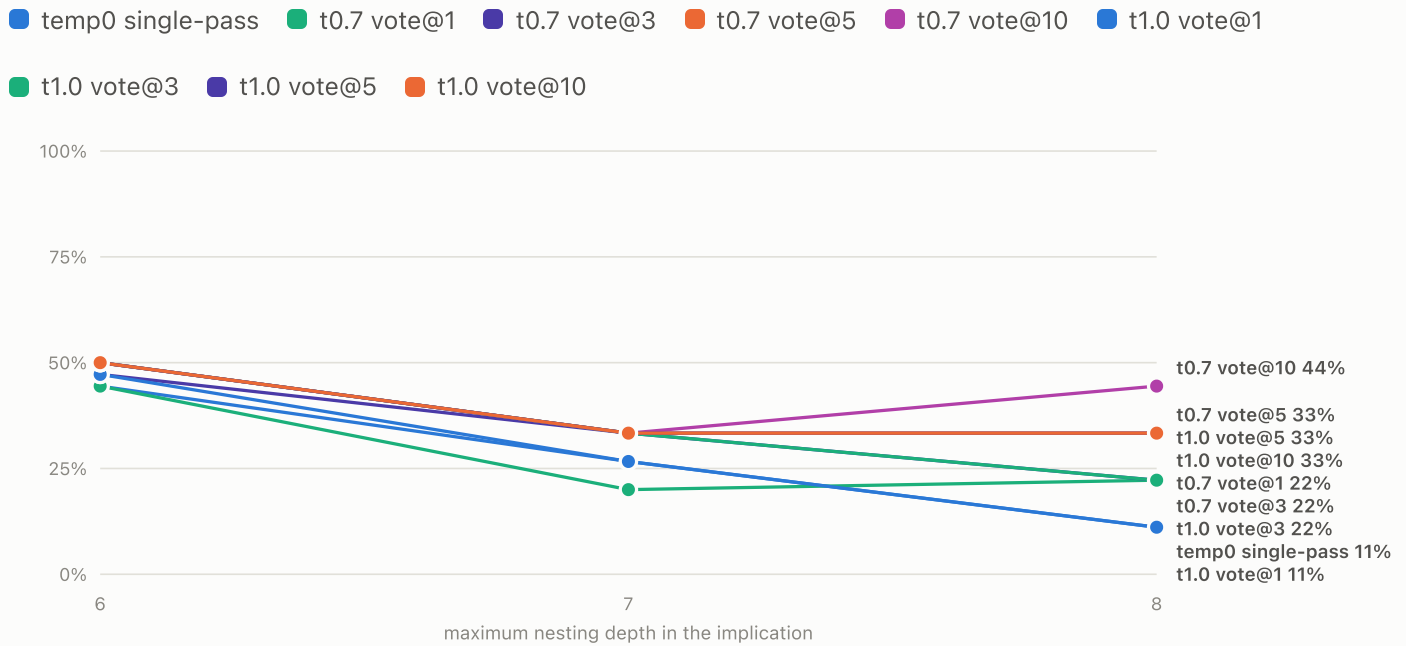
t1.0 vote@10



► Data: correct rate per model and condition

FIGURE 2 **Accuracy vs nesting depth — balanced 20**

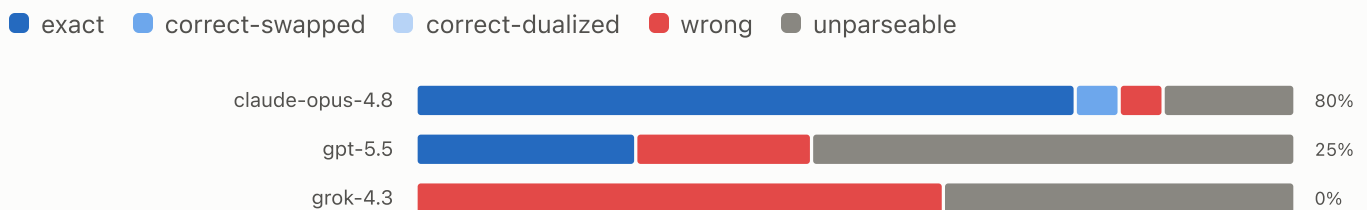
Correct rate pooled over all models, by the deepest operation nesting across the four equation sides. Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.



► Data: pooled correct rate per depth bin

FIGURE 3 **Verdict composition — temp0 single-pass · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized). Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

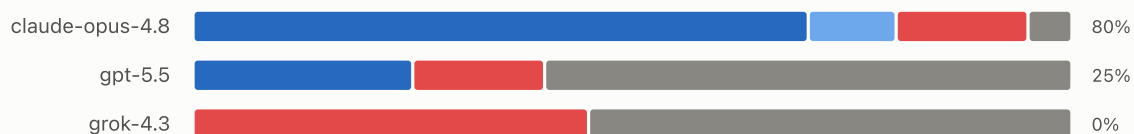


► Data: verdict counts per model

FIGURE 4 **Verdict composition — t0.7 vote@1 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 5 **Verdict composition — t0.7 vote@3 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 6 **Verdict composition — t0.7 vote@5 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 7 **Verdict composition — t0.7 vote@10 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable

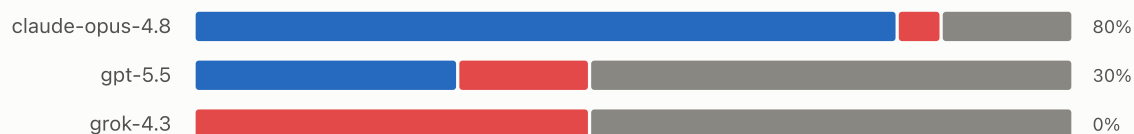


► Data: verdict counts per model

FIGURE 8 **Verdict composition — t1.0 vote@1 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 9 **Verdict composition — t1.0 vote@3 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).  
Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 10 **Verdict composition — t1.0 vote@5 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model

FIGURE 11 **Verdict composition — t1.0 vote@10 · balanced 20**

How each model's answers grade out; the right-hand figure is the correct rate (exact + swapped + dualized).

Sample: 20 pairs, 20 per per-equation operation bin 10–10 — both laws of a pair carry the bin's exact operation count.

■ exact ■ correct-swapped ■ correct-dualized ■ wrong ■ unparseable



► Data: verdict counts per model